

Highlights

A Reproducible Computational Framework for Fermion Pair Production in Pulsed Electric Fields

Othman H. Y. Zalloum

- Evolves Dirac modes and quantum-kinetic equations independently.
- Benchmarks both solvers against the exact temporal-Sauter spectrum.
- Reports norm, invariant, pulse-tail, and cross-formulation diagnostics.
- Maps absolute finite-window error over momentum and tail extent.
- Includes tested code, reproducible data, figures, and a notebook.

A Reproducible Computational Framework for Fermion Pair Production in Pulsed Electric Fields

Othman H. Y. Zalloum^{a,*}

^a*Palestine Polytechnic University, Department of Applied Mathematics and Physics, Hebron, Palestine*

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ABSTRACT

We present PAIRPULSE, a reproducible computational program for momentum-resolved fermion pair production in prescribed, spatially homogeneous electric fields in one spatial dimension. For each canonical momentum, it independently evolves a two-component Dirac mode and the equivalent non-Markovian quantum-kinetic system. The exact asymptotic occupation for a temporal Sauter pulse supplies an external benchmark, while the Dirac norm and quantum-kinetic Bloch-vector invariant provide independent integration diagnostics. The program supports superpositions of temporally displaced Sauter pulses and user-defined pulse objects through a compact field-potential interface. For $E_0/m^2 = 0.3$ and $m\tau = 2$, both solvers reproduce the analytic occupation on 41 momentum points with maximum absolute error below 8×10^{-13} . A momentum-window map shows the absolute finite-window discrepancy across the benchmark grid. A separated two-pulse calculation demonstrates the workflow for a field without a closed-form spectrum. The contribution is a transparent reference implementation with tests, data, and figure-generation scripts; no new pair-production mechanism or analytic result is claimed.

1. Program summary

Program title: PAIRPULSE

CPC library link to program files: To be assigned after acceptance.

Developer's repository link: To be assigned before public release.

Code Ocean capsule: Not applicable.

Licensing provisions: MIT License.

Programming language: Python 3.10 or later.

External routines/libraries: NumPy and SciPy; Matplotlib for plots.

Nature of problem: Momentum-resolved fermion pair occupation in a prescribed, spatially homogeneous, time-dependent electric field in 1+1 dimensions.

Solution method: Adaptive DOP853 integration of two equivalent but independently implemented state systems: the two-component Dirac equation and the non-Markovian quantum-kinetic equations. The exact temporal-Sauter solution, Dirac norm, quantum-kinetic invariant, and cross-formulation agreement are used for validation.

Restrictions: One fermion species in one spatial dimension. Spatial gradients, transverse momentum, collisions, back-reaction, photon dynamics, and self-consistent Maxwell evolution are not included.

Unusual features: The exact benchmark and two state representations are combined with independent conservation checks. Scripts regenerate the manuscript figures, tables, and run metadata.

1. Introduction

Nonperturbative pair creation by electric fields is a foundational prediction of quantum electrodynamics. Its early development includes the effective-action treatment of vacuum polarization, Dirac-state transmission in a uniform field, and pair creation in constant or alternating backgrounds [1, 2, 3, 4]. Time-dependent mean-field and kinetic descriptions subsequently clarified how pair occupations evolve, how Pauli blocking enters, and when memory effects matter [5, 6, 7, 8, 9].

For pulsed fields, the final momentum spectrum depends on the full time history of the background. Non-Markovian quantum-kinetic calculations have been compared with scattering and effective-action approaches, and the temporal Sauter field provides an especially useful exact reference case [10, 11, 13, 14, 16, 23]. Short pulses and subcycle structure produce characteristic momentum signatures, while semiclassical turning-point analyses explain interference patterns in time-dependent backgrounds [12, 15, 17, 18, 19, 22]. These studies make clear why a numerical implementation should specify its in/out convention, integration window, and momentum variable rather than report a spectrum without those choices.

Several complementary computational approaches address broader field configurations and observables. Reviews summarize the strong-field and quantum-kinetic setting; real-time lattice simulations treat dynamical fermion production; and later work examines pulse-shape effects, spatially varying fields, and the validity of the spatially uniform approximation [20, 21, 24, 25, 26, 27, 28, 29]. The definition of particle number during a field pulse also depends on the chosen basis, a

*Corresponding author

✉ ozalloum@ppu.edu (O.H.Y. Zalloum)

🌐 <https://orcid.org/0000-0001-7282-1332> (O.H.Y. Zalloum)

point clarified by analyses of adiabatic particle number [30]. This motivates a narrowly specified benchmark calculation rather than an unqualified claim of general applicability.

This paper introduces PAIRPULSE, a compact computational-physics program that evolves two independently represented mode systems under the same prescribed field: a two-component Dirac spinor and the non-Markovian quantum-kinetic variables. The exact asymptotic Sauter spectrum provides an external benchmark; conservation diagnostics test each integration internally; and a two-pulse example checks cross-formulation agreement beyond the exactly soluble case. The package includes the source, automated tests, a reproduction notebook, generated figures, machine-readable data, and run metadata. It is intended as a transparent reference implementation for method checks and modest parameter studies in the spatially homogeneous model, and it does not claim new pair-production physics.

The remainder of the paper defines the field and mode conventions, describes the software and numerical method, reports validation results, and states the scope and limitations of the program.

2. Physical model

We use natural units, $\hbar = c = 1$, and temporal gauge, $A^0 = 0$. The positive charge magnitude is denoted by q ; reversing the charge is equivalent to reflecting canonical momentum for the backgrounds considered here. For a spatially uniform vector potential $A(t)$, the kinetic momentum of a mode with canonical momentum p is

$$\pi_p(t) = p - qA(t), \quad E(t) = -\dot{A}(t). \quad (1)$$

The default pulse is

$$\begin{aligned} E(t) &= E_0 \operatorname{sech}^2\left(\frac{t-t_c}{\tau}\right), \\ A(t) &= -E_0\tau \tanh\left(\frac{t-t_c}{\tau}\right). \end{aligned} \quad (2)$$

with $E_0 \geq 0$. A pulse train is formed by summing the fields and the corresponding potentials. This preserves Eq. (1) without numerical integration of the electric field.

2.1. Dirac-mode evolution

After reduction to a two-component mode, the Hamiltonian can be written

$$H_p(t) = \pi_p(t)\sigma_1 + m\sigma_3, \quad i\partial_t\psi_p(t) = H_p(t)\psi_p(t), \quad (3)$$

where $m > 0$ and σ_i are Pauli matrices. At the initial and final times, where the pulse tails are small, the instantaneous negative- and positive-energy eigenvectors

define the in-vacuum and the out-particle state. The occupation is

$$f_p = \left| u_{p,+}^\dagger(t_f)\psi_p(t_f) \right|^2, \quad (4)$$

with $\psi_p(t_i) = u_{p,-}(t_i)$. Exact evolution preserves $\psi_p^\dagger\psi_p = 1$.

2.2. Quantum-kinetic evolution

The second formulation evolves the distribution f_p and two auxiliary functions u_p, v_p . Defining $\omega_p(t) = \sqrt{m^2 + \pi_p(t)^2}$ and $W_p(t) = qE(t)m/\omega_p(t)^2$, the equations are

$$\begin{aligned} \dot{f}_p &= \frac{1}{2}W_p u_p, \\ \dot{u}_p &= W_p(1 - 2f_p) - 2\omega_p v_p, \\ \dot{v}_p &= 2\omega_p u_p, \end{aligned} \quad (5)$$

with $(f_p, u_p, v_p) = (0, 0, 0)$ before the pulse. These equations preserve

$$(1 - 2f_p)^2 + u_p^2 + v_p^2 = 1, \quad (6)$$

which is used as an independent integration diagnostic. The asymptotic value of f_p is compared directly with Eq. (4).

2.3. Exact Sauter benchmark

For Eq. (2), let $\lambda = qE_0\tau^2$ and

$$\omega_{\pm} = \sqrt{m^2 + (p \pm qE_0\tau)^2}. \quad (7)$$

The asymptotic occupation used by the program is

$$f_p^{\text{exact}} = \frac{\cosh(2\pi\lambda) - \cosh[\pi\tau(\omega_+ - \omega_-)]}{2\sinh(\pi\tau\omega_+)\sinh(\pi\tau\omega_-)}. \quad (8)$$

The implementation evaluates an algebraically equivalent logarithmic expression to avoid overflow and underflow in the hyperbolic functions. Equation (8) is used only for a single Sauter pulse; pulse-train spectra are computed by time evolution.

3. Software and numerical method

PAIRPULSE is a Python package built around a small pulse interface exposing $E(t)$, $A(t)$, and a finite integration window. It provides an analytic Sauter pulse and finite pulse trains, but the solvers accept user-defined pulse objects implementing the same interface. This separates the field model from the mode evolution and lets users replace the pulse without changing either solver.

Both initial-value problems use SciPy's DOP853 solver through `solve_ivp`. The default relative and absolute tolerances are 2×10^{-10} and 2×10^{-12} . The integration window extends 12 duration parameters beyond the outermost pulse centers by default. A

phase-based maximum step is set using a sampled estimate of the largest instantaneous mode frequency. The window length is user-adjustable; the manuscript's tail study makes its effect explicit. The same time interval is used for both formulations.

For a single Sauter pulse, PAIRPULSE reports the exact occupation, direct Dirac occupation, quantum-kinetic occupation, norm/invariant errors, and ODE function-evaluation counts. A separate script regenerates all manuscript plots, comma-separated tables, and environment metadata. The accompanying notebook shows the central benchmark and pulse-train example. The tests cover the exact benchmark, the quantum-kinetic invariant, the zero-field limit, and pulse superposition.

Figures 4 and 5 summarize the public package contents and the sequence used to regenerate its datasets and publication figures. Both diagrams are generated by the included Python plotting script. The separate finite-window scan writes its underlying momentum-tail-extent values to a CSV file, in addition to the spectra, convergence data, and run metadata produced by the main reproduction script.

The one-dimensional momentum-integrated density is defined as

$$n_{1D} = \int \frac{dp}{2\pi} f_p, \quad (9)$$

and is evaluated with the trapezoidal rule on the supplied finite momentum grid. It is a one-dimensional density for one mode species, not a three-dimensional number density. Users must check momentum-window and grid convergence for applications beyond the examples below.

4. Validation and examples

4.1. Single Sauter pulse

The reference case uses $m = q = 1$, $E_0 = 0.3$, $\tau = 2$, and 41 equally spaced canonical momenta on $[-1.5, 1.5]$. Figure 1 compares both numerical formulations with Eq. (8). Across this grid, the largest absolute errors are 7.49×10^{-13} for the Dirac modes and 7.54×10^{-13} for the quantum-kinetic calculation. The maximum relative errors over modes with $f_p^{\text{exact}} > 10^{-12}$ are 1.69×10^{-8} and 1.69×10^{-8} , respectively. The maximum Dirac norm error is 6.24×10^{-13} ; the maximum deviation of the quantum-kinetic invariant from unity is 9.13×10^{-13} .

Finite pulse tails give a larger error than the ODE tolerance when the integration window is short. At $p = 0$ the exact value is $8.11991615993 \times 10^{-4}$. Table 1 shows that extending the window from six to twelve pulse durations reduces the absolute error by more than five orders of magnitude. The 14τ values approach the ODE error floor and are not interpreted as a general accuracy guarantee.

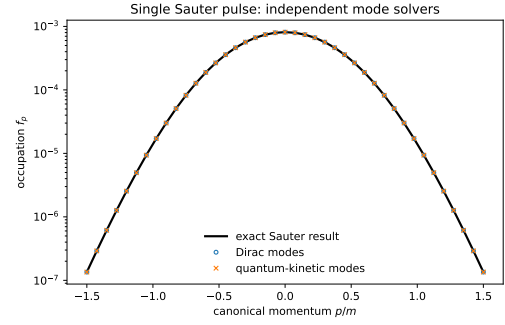


Figure 1: Single-pulse occupation spectrum for $E_0 = 0.3$, $\tau = 2$, and $m = q = 1$. The Dirac-mode and quantum-kinetic results lie on the exact Sauter curve at the plot's resolution.

4.2. Separated pulses

As a case without a closed-form benchmark, we superpose two same-polarity Sauter pulses, each with $E_0 = 0.3$ and $\tau = 1.1$, centered at $t_c = \pm 3.5$. Figure 2 shows the computed momentum spectrum on $[-1.8, 1.8]$. The two independent state representations agree over the full grid; both resolve shoulders and modulations away from the central maximum. Such structures are familiar consequences of coherent production in multiple pulses and are included here as a code demonstration, not as a new physical prediction.

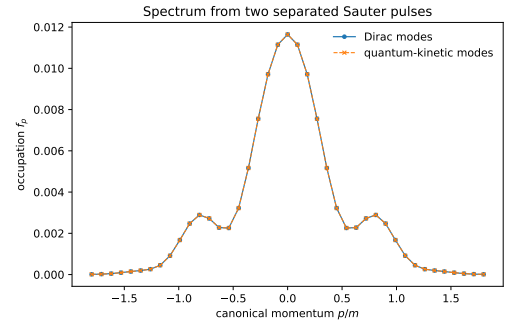


Figure 2: Occupation spectrum for two separated, same-polarity Sauter pulses. Parameters are $E_0 = 0.3$, $\tau = 1.1$, and centers $t_c = \pm 3.5$, with $m = q = 1$.

The largest absolute difference between the Dirac and quantum-kinetic occupations on this grid is 1.18×10^{-13} . For orientation, trapezoidal integration over the stated finite windows gives $n_{1D} = 1.13951 \times 10^{-4}$ for the single-pulse exact spectrum and 1.69199×10^{-3} for the double-pulse spectrum. These values are grid- and window-dependent examples; the software does not treat them as converged continuum yields without a separate momentum-domain study.

4.3. Evolution diagnostics

Figure 3 displays the absolute quantum-kinetic invariant error for the single-pulse momentum grid. The maximum error is below 10^{-12} in this run. The invariant is a useful internal diagnostic, but it does

Table 1

Finite-window convergence at $p = 0$ for the single Sauter benchmark. Errors are absolute differences from Eq. (8).

Tail half-width L/τ	Exact	Dirac absolute error	QKE absolute error
6	$8.11991615993 \times 10^{-4}$	1.1762×10^{-7}	1.1762×10^{-7}
8	$8.11991615993 \times 10^{-4}$	2.1989×10^{-9}	2.1989×10^{-9}
10	$8.11991615993 \times 10^{-4}$	4.0728×10^{-11}	4.0721×10^{-11}
12	$8.11991615993 \times 10^{-4}$	7.4813×10^{-13}	7.5435×10^{-13}
14	$8.11991615993 \times 10^{-4}$	1.2825×10^{-14}	6.7149×10^{-15}

not replace comparison with the exact occupation or
a convergence study in momentum and time.

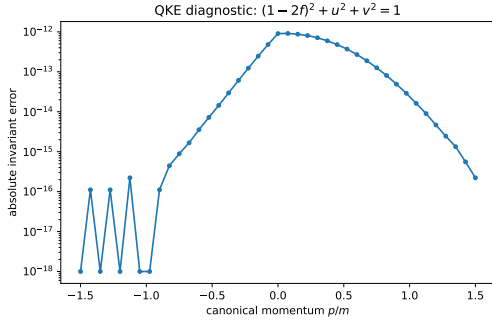


Figure 3: Absolute deviation of $(1 - 2f_p)^2 + u_p^2 + v_p^2$ from unity after the single-pulse evolution.

Short integration windows can shift the computed spectrum away from its asymptotic value. Figure 6 shows the absolute error for both formulations. The scan uses 41 momenta and six tail half-widths, with L/τ from 4 to 14 in steps of 2. Parameters are $E_0/m^2 = 0.3$ and $m\tau = 2$.

Here L is the half-width from the pulse center to either integration endpoint. The plotted discrepancy includes finite-window truncation and fixed-tolerance integration error; values near the solver floor are not pure tail error. The CSV records both occupations, the exact result, errors, and solver diagnostics. Because the field is spatially homogeneous, the map axes are momentum and window extent.

5. Limitations and intended use

The implementation assumes a prescribed, spatially homogeneous background and one spatial dimension. It does not include back-reaction, collisions, spatial gradients, transverse momentum, scalar particles, or a self-consistent electromagnetic field. This restriction is physically consequential: calculations beyond the dipole or spatially uniform approximation show that field variation can change pair yields and momentum spectra [25, 27, 28, 29]. PAIRPULSE is therefore a reference implementation for the stated homogeneous model, not a substitute for lattice, Wigner, or spatially resolved calculations when these effects matter.

The exact Sauter formula validates asymptotic occupation for a single pulse; it does not validate every user-supplied field. For new profiles, users should vary tolerances, pulse-tail windows, time-step limits, and momentum resolution, and compare physical observables with an independent method where possible. Particle-number interpretation during an active pulse requires particular care because instantaneous occupation can be basis dependent [30].

The code emphasizes clarity and cross-validation over throughput. Each momentum mode is integrated independently in Python, so the present package is most appropriate for reference calculations, method checks, and modest parameter sweeps. Parallel execution, compiled kernels, and large-scale parameter scans are outside the current release.

6. Conclusion

PAIRPULSE provides an inspectable implementation of two established formulations for fermion pair creation in homogeneous pulsed electric fields. The exact Sauter solution, norm and invariant diagnostics, pulse-tail convergence data, separated-pulse example, and executable notebook make the numerical conventions reproducible. The two numerical formulations reproduce the exact benchmark to below 8×10^{-13} absolute error for the stated parameter set and agree to about 1.2×10^{-13} for the two-pulse example. These results validate the implementation for the tested cases; they do not establish a general accuracy guarantee or claim new pair-production physics.

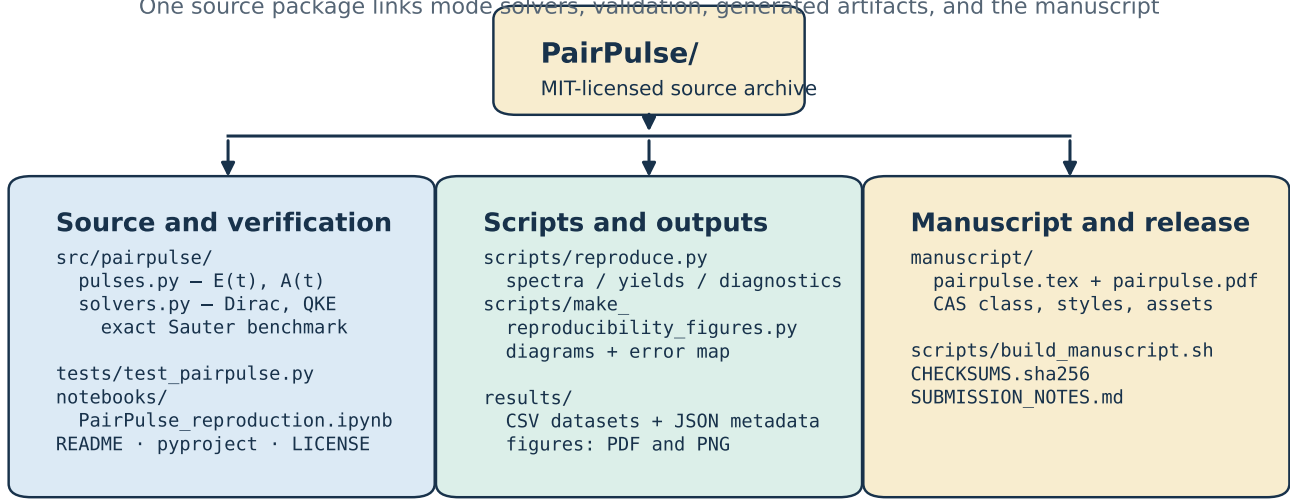
The deliberately limited model makes the package useful as a reference for homogeneous-field calculations and as a base for extensions. Applications involving spatial dependence, back-reaction, or additional particle degrees of freedom require further derivations and validation against appropriate independent methods.

Code and data availability

The supplementary reproducibility package supplied with this manuscript contains the MIT-licensed Python code, unit tests, reproduction script, notebook, generated figures, CSV tables, run metadata, and the CAS class and style files used to build the manuscript.

PairPulse public reproducibility package

One source package links code, solvers, validation, generated artifacts, and the manuscript



Rebuild: run both Python scripts → regenerate the CAS PDF → verify CHECKSUMS.sha256

Generated datasets and figures are included alongside their source-generating code.

Figure 4: Structure of the public PAIRPULSE package. Source modules, tests, notebook, and run scripts are shown alongside the generated datasets, figures, manuscript, and release checks. The diagram is produced in Python from the package's declared file structure.

A public repository or archive identifier can be added when the submission package is formally released.

The corresponding author is responsible for reviewing the final manuscript and checking its scientific content, references, and reported results before submission.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

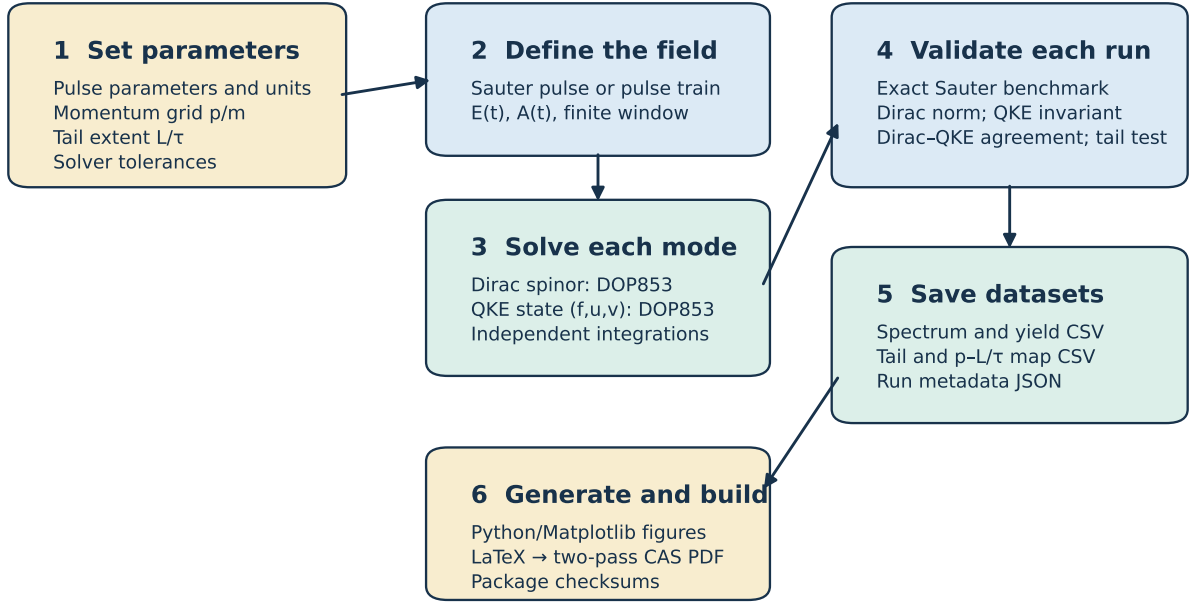
ChatGPT, an AI-assisted language model developed by OpenAI, was used to assist with literature searching, manuscript drafting, and LaTeX formatting.

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Computational workflow for the reproducibility datasets and figures

The same field and momentum conventions are used by both independent mode formulations



Scripts: `reproduce.py · make_reproducibility_figures.py · build_manuscript.sh`

Figure 5: Computational workflow used to generate the reproducibility datasets and publication figures. Each momentum mode is evolved independently with the Dirac and quantum-kinetic systems; the exact Sauter result and conservation diagnostics are then used where applicable before data export, plotting, and manuscript compilation.

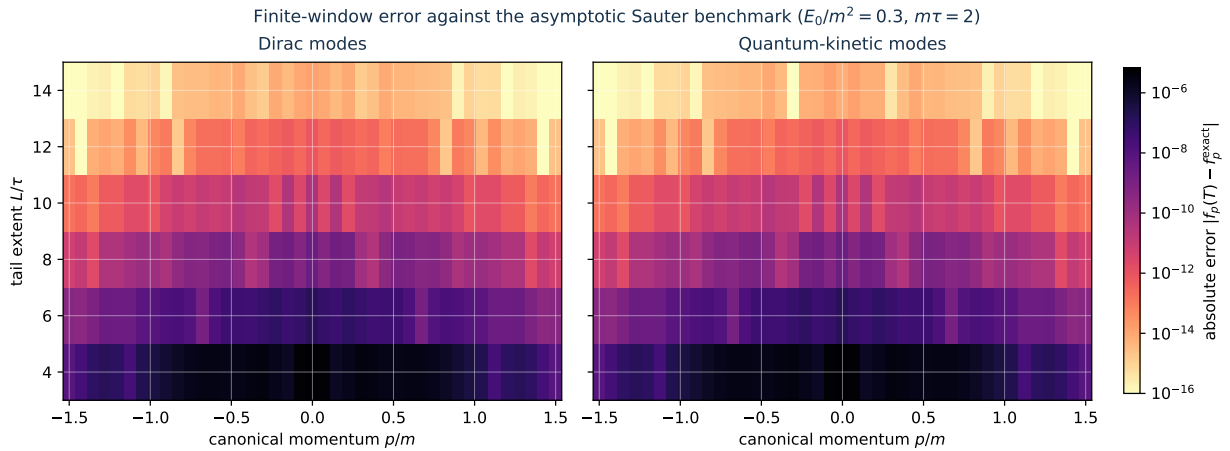


Figure 6: Absolute finite-window discrepancy $|f_p(L) - f_p^{\text{exact}}|$ from the asymptotic Sauter benchmark, shown on a logarithmic color scale for the Dirac and quantum-kinetic formulations. The horizontal coordinate is canonical momentum and the vertical coordinate is the tail half-width L/τ ; integration endpoints are $t_c \pm L$. The field is spatially uniform, so this is not a spatial error distribution.

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